AirPulse: A Context-Aware Health Risk Assessment System Using Smartphone-Based PPG and Air Quality Integration

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***Abstract*—This paper presents AirPulse, a context-aware health monitoring system integrating smartphone-based photoplethysmography (PPG), real-time Air Quality Index (AQI), and user health profiles. The system employs Fast Fourier Transform (FFT) for heart rate estimation and a rule-based model for interpretable risk classification. A user study involving 42 participants and real-world testing demonstrates the system's usability and effectiveness. The proposed system bridges the gap between physiological sensing and environmental awareness.**

I. Introduction

Cardiovascular diseases are influenced by both physiological conditions and environmental factors such as air pollution. Existing health monitoring systems typically operate in isolation, focusing on either heart rate or AQI independently. This results in incomplete insights and poor decision-making.

AirPulse integrates multimodal data streams, including PPG-based heart rate, AQI, and personal health attributes, to provide a unified, context-aware risk assessment. This approach aligns with the principles of ubiquitous computing and personalized healthcare.

II. Related Work

Photoplethysmography (PPG) has been widely used for non-invasive heart rate monitoring [1]. Frequency-domain approaches using FFT have been shown to be robust against noise [2].

Air pollution has been strongly linked to cardiovascular diseases, particularly due to $PM_{2.5}$ exposure [3]. Studies indicate increased risks of hypertension, arrhythmias, and cardiac mortality [4].

Wearable health monitoring systems have been explored extensively [5], but they often require specialized hardware. Recent works in context-aware systems emphasize multimodal data fusion [6], yet few systems provide interpretable outputs for users.

AirPulse differentiates itself by combining physiological sensing, environmental awareness, and user profiling into a single, accessible mobile platform.

III. Methodology

The AirPulse system follows a structured signal processing pipeline for extracting heart rate from smartphone-based photoplethysmography (PPG) and integrating it with environmental and user data for risk assessment.

A. Signal Acquisition (PPG Capture)

The system utilizes the smartphone camera and flashlight to capture photoplethysmographic signals. When the user places their fingertip over the camera lens, the flashlight illuminates the skin, and the camera records subtle variations in light intensity caused by pulsatile blood flow. These variations correspond to changes in blood volume during each cardiac cycle. The captured video frames act as the raw physiological signal source.

B. Signal Extraction and Pre-processing

The captured video frames are processed to extract a usable PPG signal. The red channel intensity is computed for each frame, as it is most sensitive to blood flow variations. This results in a temporal signal:

$$x_{red}(t) = \frac{1}{N}\sum_{i=1}^{N} R_i(t) \tag{1}$$

where $R_i(t)$ represents the red channel intensity of each pixel and N is the total number of pixels.

To improve signal quality, noise reduction and smoothing techniques are applied to remove disturbances caused by finger movement, lighting inconsistencies, and sensor noise. A band-pass filter is then applied in the range of 0.7–4 Hz to retain only the frequency components relevant to human heart rate:

$$x_{filtered}(t) = BP(x_{red}(t)) \tag{2}$$

C. Frequency Domain Transformation

After obtaining a clean PPG waveform, the signal is transformed into the frequency domain using the Fast Fourier Transform (FFT):

$$X(f) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi fn/N} \tag{3}$$

The FFT decomposes the signal into its constituent frequency components, producing a power spectrum. The dominant peak in the spectrum corresponds to the heart rate frequency. This approach is more robust than time-domain peak detection, especially in noisy conditions.

D. Heart Rate Estimation

The dominant frequency f_{peak} is identified from the power spectrum and converted into beats per minute (BPM):

$$HR = 60 \times f_{peak} \tag{4}$$

The computed heart rate is displayed to the user and passed to the risk assessment module as a key physiological parameter.

E. Integration with AQI and User Profile

The estimated heart rate is combined with real-time AQI data obtained through location-based APIs and user-specific attributes such as age, exercise frequency, and medical conditions. These inputs are fed into a rule-based risk scoring model to generate a final health risk classification.

This complete pipeline ensures reliable physiological signal extraction, robust heart rate estimation, and meaningful integration with environmental and personal health data.

IV. Risk Model

$$RiskScore = S_{HR} + S_{AQI} + S_{Resp} + S_{Cardio} + S_{Age} + S_{Exercise} \tag{5}$$

Risk Levels: Low (0–3), Moderate (4–6), High (7–10)

V. User Workflow

The system follows a structured pipeline:

1. Profile Setup: User inputs age, exercise frequency, and medical conditions.

2. Dashboard: Displays real-time HR, AQI, and risk level.

3. Measurement: Camera-based PPG captures heart rate.

4. AQI Fetching: Location-based API retrieves environmental data.

5. Risk Assessment: Combined model outputs risk category.

6. Recommendations: System provides actionable health suggestions.

VI. Survey and User Study

A usability survey was conducted with 42 participants to evaluate the system.

A. User Experience Results

- 95.2
- 92.9
- 70

B. Graphical Analysis

The survey data was visualized using bar charts and distribution plots.

Graph 1: UI Simplicity Distribution A bar graph showed that the majority of users strongly agreed or agreed that the interface is easy to use.

Graph 2: Measurement Comfort A distribution graph indicated that most users experienced no discomfort, validating the usability of camera-based sensing.

Graph 3: Risk Interpretation Preference Users showed a strong preference for categorical outputs (Low/Moderate/High) over raw numbers, highlighting the importance of interpretability.

These graphical insights confirm that AirPulse successfully reduces cognitive load and improves decision-making.

VII. Experimental Results

The system was evaluated under different conditions:

- Accuracy: ± 4–7 BPM at rest
- Accuracy: ± 8–12 BPM under motion
- AQI data matched API values accurately

The risk model demonstrated logical consistency: Higher HR + Higher AQI $\rightarrow$ Higher Risk

VIII. Discussion

The integration of multimodal data enhances contextual awareness. Unlike traditional systems, AirPulse provides interpretable insights, which improves user trust and usability. The survey results further validate the importance of simplicity and actionable feedback.

IX. Limitations

- PPG sensitivity to motion and lighting
- Rule-based model lacks adaptability
- AQI may not reflect micro-environments
- Not a clinical diagnostic system

X. Future Work

Future work includes integrating activity recognition using accelerometers, adopting machine learning models for prediction, and improving signal robustness.

XI. Conclusion

AirPulse demonstrates the effectiveness of combining physiological sensing and environmental awareness for health monitoring. The system provides accurate, interpretable, and actionable insights, making it suitable for real-world deployment.

References

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